第四次作业

1. 考虑极坐标系下 $r\sim r+dr$, $\theta\sim\theta+d\theta$ 一段微元的受力. 设振幅为 $u(r,\theta,t)$

如作出图中所示, $\sin\alpha_2 = \frac{1}{dr}\left(u(r+dr,\theta+d\theta,t)-u(r,\theta+d\theta,t)\right)$

$= \frac{\partial u}{\partial r}(r,\theta+d\theta,t)$ $\quad\quad \sin\alpha_1 = \frac{\partial u}{\partial r}(r,\theta,t)$

径向受力: $F_r = T(r+dr)\cdot d\theta\,\frac{\partial u}{\partial r}(r+dr,\theta,t) - Tr d\theta\,\frac{\partial u}{\partial r}(r,d\theta,t)$

$= T dr d\theta\,\frac{\partial u}{\partial r}(r,\theta,t) + Tr dr d\theta\,\frac{\partial^2 u}{\partial r^2}(r,\theta,t)$

切向受力: 由图, $\sin\beta = \frac{u(r,\theta+d\theta,t)-u(r,\theta,t)}{r d\theta} = \frac{1}{r}\frac{\partial u}{\partial\theta}(r,\theta,t)$

$F_\theta = T dr\left(\frac{1}{r}\frac{\partial u}{\partial\theta}(r,\theta+d\theta,t) - \frac{1}{r}\frac{\partial u}{\partial\theta}(r,\theta,t)\right) = \frac{1}{r}T dr d\theta\,\frac{\partial^2 u}{\partial\theta^2}(r,\theta,t)$

由受力, $dm\cdot u_{tt} = \rho_m dr d\theta\, u_{tt} = F_r + F_\theta = T dr d\theta\left(u_r + r u_{rr} + \frac{1}{r}u_{\theta\theta}\right)$

$\Rightarrow \frac{\rho_m}{T}u_{tt} = u_{rr} + r u_{rr} + \frac{1}{r}u_{\theta\theta}$

换回直角坐标系: $\frac{\rho_m}{T}u_{tt} = u_{xx} + u_{yy}$.

2. $$\begin{cases} u_t = a^2 u_{xx} - b^2 u & (0<x<L,\ t>0) \\ u|_{t=0} = \sqrt{2}\, C \cos\frac{\pi}{4L}x & (0\le x\le L) \\ u_x|_{x=0} = 0. \quad u|_{x=L} = C. & (t>0) \end{cases}$$

记 $v = u - C$ $\Rightarrow$ $$\begin{cases} v_t = a^2 v_{xx} - b^2 v \\ v|_{t=0} = \sqrt{2}\, C\cos\frac{\pi}{4L}x - C \\ v_x|_{x=0} = 0. \quad v|_{x=L} = 0. \end{cases}$$

此时有正交完备基 $\{1, \sqrt{2}\cos\frac{1}{2}\frac{x}{L}\pi, \cdots, \sqrt{2}\cos(n-\frac{1}{2})\frac{x}{L}\pi, \cdots\}$

记 $v = \sum_{n=1}^{+\infty} C_n(t)\cdot\cos(n-\frac{1}{2})\frac{x}{L}\pi$. $C_n(t)$满足:

$$\frac{dC_n}{dt} = \left(-a^2(n-\tfrac{1}{2})^2\frac{\pi^2}{L^2} - b^2\right)C_n \Rightarrow C_n(t) = D_n\cdot\exp\left(-\left(a^2(n-\tfrac{1}{2})^2\frac{\pi^2}{L^2} + b^2\right)t\right)$$

$D_n$由初始条件确定: $D_n = \frac{2}{L}\int_0^L \sqrt{2}\, C\cos\frac{\pi}{4L}x\cdot\cos(n-\frac{1}{2})\pi\frac{x}{L}\, dx$

$$= \frac{1}{L}\sqrt{2}\, C\int_0^L \left(\cos\pi\frac{x}{L}(n-\tfrac{1}{4}) + \cos\pi\frac{x}{L}(n-\tfrac{3}{4})\right) dx$$

$$= \frac{\sqrt{2}}{L} C\left(\frac{L}{(n-1/4)\pi}\sin((n-\tfrac{1}{4})\pi) + \frac{L}{(n-3/4)\pi}\sin((n-\tfrac{3}{4})\pi)\right).$$

$$= (-1)^{n+1}\frac{C}{\pi}\frac{2n-1}{(n-1/4)(n-3/4)}$$

故解为：$u(x,t)=\sum_{n=1}^{+\infty}(-1)^{n+1}\frac{2n-1}{(n-1/4)(n-3/4)}\frac{C}{\pi}e^{-(a^2(n-\frac{1}{2})^2\frac{\pi^2}{l^2}+b^2)t}\cdot\cos(n-\frac{1}{2})\pi\frac{x}{l}+C.$

3.

(1) $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(2x+\lambda)y=0$

$\Rightarrow x\frac{d}{dx}(x\frac{dy}{dx})+(2x+\lambda)y=0 \Rightarrow \frac{d}{dx}(x\frac{dy}{dx})+2y+\lambda\cdot\frac{1}{x}y=0$

(2) $x(1-x)\frac{d^2y}{dx^2}+(a-bx)\frac{dy}{dx}-\lambda y=0$

$\Rightarrow x(1-x)\cdot\frac{(1-x)^{a-b}}{x^b}\frac{d}{dx}\left(\frac{x^b}{(1-x)^{a-b}}\cdot\frac{dy}{dx}\right)-\lambda y=0.$

4. $$\begin{cases} X''+\lambda X=0 \\ a_1X(0)+\beta_1X'(0)=0 \\ a_2X(l)+\beta_2X'(l)=0 \end{cases} \Rightarrow X = C_1\sin(\sqrt{\lambda}x)+C_2\cos(\sqrt{\lambda}x)$$

$$\Rightarrow \begin{cases} a_1C_2+\beta_1C_1\sqrt{\lambda}=0 \\ a_2C_1\sin(\sqrt{\lambda}l)+a_2C_2\cos(\sqrt{\lambda}l)+C_1\beta_2\sqrt{\lambda}\cos(\sqrt{\lambda}l)-C_2\beta_2\sqrt{\lambda}\sin(\sqrt{\lambda}l)=0 \end{cases}$$

若需方程有非零解，则需有:

$$\begin{vmatrix} \beta_1\sqrt{\lambda} & a_1 \\ a_2\sin(\sqrt{\lambda}l)+\beta_2\sqrt{\lambda}\cos(\sqrt{\lambda}l) & a_2\cos(\sqrt{\lambda}l)-\beta_2\sqrt{\lambda}\sin(\sqrt{\lambda}l) \end{vmatrix}=0$$

$$\Rightarrow \beta_1a_2\sqrt{\lambda}\cos(\sqrt{\lambda}l)-\beta_1\beta_2\lambda\sin(\sqrt{\lambda}l)=a_1a_2\sin(\sqrt{\lambda}l)+a_1\beta_2\sqrt{\lambda}\cos(\sqrt{\lambda}l)$$

$$\Rightarrow \tan(\sqrt{\lambda_n}l)=\frac{a_2\beta_1-a_1\beta_2}{a_1a_2+\beta_1\beta_2\lambda_n}\sqrt{\lambda_n}. \quad n=0,1,2,\cdots$$

$$\sqrt{\lambda_n} \in \left(\frac{n-\frac{1}{2}}{l}\pi,\ \frac{n+\frac{1}{2}}{l}\pi\right).\quad X=\sum_{n=0}^{+\infty}\left(\sin(\sqrt{\lambda_n}x)-\frac{\beta_1}{a_1}\sqrt{\lambda_n}\cos(\sqrt{\lambda_n}x)\right)\cdot C_n$$

正交性: $\forall \lambda_n, \lambda_m, n\neq m.$

$$\int_0^l\left(\sin(\sqrt{\lambda_n}x)-\frac{\beta_1}{a_1}\sqrt{\lambda_n}\cos(\sqrt{\lambda_n}x)\right)\left(\sin(\sqrt{\lambda_m}x)-\frac{\beta_1}{a_1}\sqrt{\lambda_m}\cos(\sqrt{\lambda_m}x)\right)dx$$

$$\int_0^l \cos\sqrt{\lambda_n}x\cos\sqrt{\lambda_m}x\,dx=\frac{1}{\sqrt{\lambda_m}}\cos\sqrt{\lambda_n}x\sin\sqrt{\lambda_m}x+\frac{\sqrt{\lambda_n}}{\sqrt{\lambda_m}}\int_0^L\sin\sqrt{\lambda_n}x\sin\sqrt{\lambda_m}x\,dx$$

$$\int_0^L \cos\sqrt{\lambda_n}x\sin\sqrt{\lambda_m}x\,dx=\frac{1}{\sqrt{\lambda_n}}\sin\sqrt{\lambda_n}x\sin\sqrt{\lambda_m}x-\frac{\sqrt{\lambda_m}}{\sqrt{\lambda_n}}\int_0^L\sin\sqrt{\lambda_n}x\cos\sqrt{\lambda_m}x\,dx$$

故原积分 $I=\left(\frac{\beta_1}{a_1}\right)^2\sqrt{\lambda_n}\cos\sqrt{\lambda_n}x\sin\sqrt{\lambda_m}x+\left(1+\left(\frac{\beta_1}{a_1}\right)^2\lambda_n\right)I_1$

$$-\frac{\beta_1}{a_1}\sin\sqrt{\lambda_n}x\sin\sqrt{\lambda_m}x$$

其中 $I_1=\int_0^L\sin\sqrt{\lambda_n}x\sin\sqrt{\lambda_m}x\,dx$

$$\Rightarrow I = \left(\frac{\beta_1}{a_1}\right)^2 \sqrt{\lambda_n} \cos\sqrt{\lambda_n}L \sin\sqrt{\lambda_m}L - \frac{\beta_1}{a_1} \sin\sqrt{\lambda_n}L \sin\sqrt{\lambda_m}L$$

$$+ \left(1 + \left(\frac{\beta_1}{a_1}\right)^2 \lambda_n\right) \cdot \frac{1}{2} \left( \frac{\sin(\sqrt{\lambda_n} - \sqrt{\lambda_m})L}{\sqrt{\lambda_n} - \sqrt{\lambda_m}} - \frac{\sin(\sqrt{\lambda_n} + \sqrt{\lambda_m})L}{\sqrt{\lambda_n} + \sqrt{\lambda_m}} \right)$$

$$= k^2 \sqrt{\lambda_n} \cos\sqrt{\lambda_n}L \sin\sqrt{\lambda_m}L - k \sin\sqrt{\lambda_n}L \sin\sqrt{\lambda_m}L \qquad \left(k = \frac{\beta_1}{a_1}\right)$$

$$+ \left(1 + \left(\frac{\beta_1}{a_1}\right)^2 \lambda_n\right) \frac{a \sin\sqrt{\lambda_n}L \cos\sqrt{\lambda_m}L - k \cos\sqrt{\lambda_n}L \sin\sqrt{\lambda_m}L}{\lambda_n - \lambda_m}$$

上式同时除以 $\cos\sqrt{\lambda_n}L \sin\sqrt{\lambda_m}L$，得：

$$I \propto k^2 \sqrt{\lambda_n} - k \frac{a_2\beta_1 - a_1\beta_2}{a_1a_2 + \beta_1\beta_2\lambda_n} \sqrt{\lambda_n} + \frac{1 + k^2\lambda_n}{\lambda_n - \lambda_m} \left( -1 + \frac{a_1a_2 + \beta_1\beta_2\lambda_m}{a_1a_2 + \beta_1\beta_2\lambda_n} \sqrt{\frac{\lambda_n}{\lambda_m}} \right)$$

当 $\lambda_n = \lambda_m = \lambda$ 时，$I = \frac{\beta_1}{a_1} \sqrt{\lambda} \sin\sqrt{\lambda}L \left( \frac{\beta_1}{a_1} \cos\sqrt{\lambda}L - \sin\sqrt{\lambda}L \right) = \|f_n\|_2^2$

$$\Rightarrow \|f_n\|_2 = \sqrt{I}$$

做法2：设 $f_n, f_m$ 为 $\lambda_n, \lambda_m$ 的特征函数

$$\begin{cases} f_n'' + \lambda_n f_n = 0 \\ f_m'' + \lambda_m f_m = 0 \end{cases}$$

$$\Rightarrow (\lambda_n - \lambda_m)\int_0^L f_n f_m \, dx = \int_0^L (f_n f_m'' - f_m f_n'')\, dx$$

$$= (f_n f_m' - f_m f_n')\Big|_0^L$$

代入 $f_n = \sin\sqrt{\lambda_n}x - \frac{\beta_1}{a_1}\sqrt{\lambda_n}\cos\sqrt{\lambda_n}x$

$f_m = \sin\sqrt{\lambda_m}x - \frac{\beta_1}{a_1}\sqrt{\lambda_m}\cos\sqrt{\lambda_m}x$

$$\Rightarrow \int_0^L f_n f_m dx \propto (\sin\sqrt{\lambda_n}L - k\sqrt{\lambda_n}\cos\sqrt{\lambda_n}L)(\sqrt{\lambda_m}\cos\sqrt{\lambda_m}L + \frac{\beta_1\lambda_m}{a_1}\sin\sqrt{\lambda_m}L)$$
$$+ (-\sin\sqrt{\lambda_m}L + k\sqrt{\lambda_m}\cos\sqrt{\lambda_m}L)(\sqrt{\lambda_n}\cos\sqrt{\lambda_n}L + \frac{\beta_1}{a_1}\lambda_n\sin\sqrt{\lambda_n}L)$$

$$= \sqrt{\lambda_m}(1+k^2)SC + k(\lambda_m - \lambda_n)SS - (k^2\lambda_m\sqrt{\lambda_n} + \sqrt{\lambda_n})CS = 0$$

5. 设 $\begin{cases} \dfrac{d}{dx}\left(p(x)\dfrac{dy_n}{dx}\right)+[\lambda_n \rho(x)-q(x)]y_n=0 & ① \\ \dfrac{d}{dx}\left(p(x)\dfrac{dy_m}{dx}\right)+[\lambda_m \rho(x)-q(x)]y_m=0 & ② \end{cases}$

则 $y_m ① - y_n ② = 0 \Rightarrow$

$$\rho(\lambda_n-\lambda_m)y_n y_m = y_n\frac{d}{dx}\left(p(x)\frac{dy_m}{dx}\right)-y_m\frac{d}{dx}\left(p(x)\frac{dy_n}{dx}\right)$$

两侧积分得:

$$(\lambda_n-\lambda_m)\int_a^b \rho y_n y_m dx=\int_a^b\left(y_n\frac{d}{dx}\left(p\frac{dy_m}{dx}\right)-y_m\frac{d}{dx}\left(p\frac{dy_n}{dx}\right)\right)dx$$

$$=\left(y_n p\frac{dy_m}{dx}-y_m p\frac{dy_n}{dx}\right)\Big|_a^b+0.$$

$$=p(b)\left(y_n(b)y_m'(b)-y_m(b)y_n'(b)\right)-p(a)\left(y_n(a)y_m'(a)-y_m(a)y_n'(a)\right)$$

代入边界条件和 $p(a)=p(b)$ 得:

$$(\lambda_n-\lambda_m)\int_a^b \rho y_n y_m dx=p(a)\left[\left(a_{11}y_n(a)+a_{12}y_n'(a)\right)\left(a_{21}y_m(a)+a_{22}y_m'(a)\right)\right.$$

$-(a_{11}y_m(a)a_{12}y_m'(a))(a_{21}y_n(a)+a_{22}y_n'(a)) - y_n y_m'(a)+y_m y_n'(a)]$

$$= p(a)\left(\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22}\end{vmatrix} - 1\right)(y_n(a)y_m'(a) - y_m(a)y_n'(a)) = 0$$

（$=0$ 由条件）

由于已经证过实函数的 S-L 问题不存在简并，因此可以断言：

$$\int_a^b \rho\, y_n y_m \, dx = 0 \quad . \square$$

6. $\begin{cases} u_t = a^2 u_{xx} + f \\ u|_{t=0} = \varphi(x) \end{cases}$ . $\mathcal{L}[u] = \hat{u}(x,p) \Rightarrow \hat{u_t} = p\hat{u} - u|_{t=0} = p\hat{u} - \varphi$

因此有：$p\hat{u} - \varphi = a^2\hat{u}_{xx} + \hat{f}$.

考虑对 $x$ 的 fourier 变换：$p\hat{u}(\omega,p) - \hat{\varphi}(\omega) = -a^2\omega^2\hat{u}(\omega,p) + \hat{f}(\omega,p)$

$$\Rightarrow \hat{u}(\omega,p) = \frac{1}{p+a^2\omega^2}\left(\hat{\varphi}(\omega) + \hat{f}(\omega,p)\right)$$

$$\Rightarrow \hat{u}(\omega,t) = e^{-a^2\omega^2 t}\cdot\hat{\varphi}(\omega) + \int_0^t \hat{f}(\omega,\tau)\cdot e^{-a^2\omega^2(t-\tau)}\,d\tau.$$

$$\Rightarrow u(x,t) = F^{-1}[\hat{u}(\omega,t)] = \varphi(x) * \left(\frac{1}{2a\sqrt{\pi t}}e^{-\frac{x^2}{4a^2t}}\right) + \sim$$

$$= \int_{-\infty}^{+\infty}\varphi(\xi)\frac{1}{2a\sqrt{\pi t}}e^{-\frac{(x-\xi)^2}{4a^2t}}\,d\xi$$

$$+ \int_0^t d\tau\int_{-\infty}^{+\infty} f(\xi,\tau)\cdot\frac{1}{2a\sqrt{\pi(t-\tau)}}e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}}\,d\xi,$$

7. $\begin{cases} u_{tt} = c^2 u_{xx} \\ u|_{t=0} = \varphi(x), \quad u_t|_{t=0} = \psi(x) \end{cases}$

先对t做Laplace变换：

$$p^2 \hat{u} - p\varphi - \psi = c^2 \hat{u}_{xx}$$

再对x做Fourier变换：

$$p^2 \hat{u} - p\hat{\varphi} - \hat{\psi} = -\omega^2 c^2 \hat{u}$$

$$\Rightarrow \hat{u}(\omega, p) = \frac{p\hat{\varphi} + \hat{\psi}}{p^2 + c^2\omega^2}.$$

先对$\hat{u}$做Laplace逆变换：

$$\hat{u}(\omega, t) = \hat{\varphi}\, \mathcal{L}^{-1}\left[\frac{p}{p^2+c^2\omega^2}\right] + \hat{\psi}\, \mathcal{L}^{-1}\left[\frac{1}{p^2+c^2\omega^2}\right]$$

$$= \frac{1}{2}\hat{\varphi}\left(e^{-ic\omega t} + e^{ic\omega t}\right) + \frac{\hat{\psi}}{2ic\omega}\left(e^{ic\omega t} - e^{-ic\omega t}\right)$$

再对$\hat{u}(\omega,t)$做Fourier逆变换.

$$F^{-1}\left[\frac{1}{2}\hat{\varphi}\left(e^{-ic\omega t}+e^{ic\omega t}\right)\right]=\frac{1}{2}\left(\varphi(x+ct)+\varphi(x-ct)\right)$$

$$F^{-1}\left[\frac{\hat{\psi}}{2ic\omega}\left(e^{ic\omega t}-e^{-ic\omega t}\right)\right]=\int_{x}^{x+ct}\frac{\psi}{2c}dx-\int_{x}^{x-ct}\frac{\psi}{2c}dx$$

$$=\int_{x-ct}^{x+ct}\frac{\psi}{2c}dx$$

$$\Rightarrow u(x,t)=\frac{1}{2}\left(\varphi(x+ct)+\varphi(x-ct)\right)+\frac{1}{2c}\int_{x-ct}^{x+ct}\psi(\xi)d\xi.$$